

Report: Active Directory Federation

1. Introduction

Active Directory Federation Services (ADFS) is a feature in Windows Server that enables secure identity federation and Single Sign-On (SSO) capabilities across trusted systems and external applications. It allows users within the internal network to access web-based applications—both on-premises and in the cloud—without repeated login prompts. ADFS achieves this by issuing security tokens containing user identity and claims information based on Active Directory authentication. In this task, the primary goal was to install and configure ADFS within the task.local domain and establish a secure federation service accessible through the internal DNS hostname adfs.task.local. We also aimed to verify the system's ability to monitor and maintain trust relationships within the federation setup. By importing a self-signed certificate and ensuring proper endpoint accessibility, we laid the groundwork for future integrations. This task did not involve external applications but instead focused on preparing the environment for secure SSO workflows and verifying the integrity and reliability of trust relationships using built-in PowerShell cmdlets and event logs. The entire setup and validation process was conducted in a local lab environment using Windows Server 2022.

2. Objective

The primary objectives of this task set were to install and configure the Active Directory Federation Services (ADFS) role on a Windows Server 2022 machine to enable secure federation services. This included setting up internal Domain Name System (DNS) records for the federation service host, specifically creating an entry for "adfs.task.local" to ensure proper name resolution within the domain. Additionally, the task required importing and binding an SSL certificate to the ADFS service to secure communications between clients and the federation server. The federation configuration was completed using the built-in ADFS setup wizard, which guided the process of establishing the federation service. Finally, the configuration and health of federation trust relationships were validated by executing relevant PowerShell cmdlets and reviewing event logs to confirm correct operation and identify any potential issues.

3. Tools used

- Windows Server 2022
- Active Directory Domain Services
- DNS Manager
- ADFS Management Console
- PowerShell
- Event Viewer

4. Methodology

4.1 Install ADFS and Configure a Test SSO Application

The ADFS role was installed on SERVER1 using Server Manager. An internal DNS A-record was created for adfs.task.local pointing to 192.168.48.137. A self-signed SSL certificate was generated, exported in .pfx format, and imported during the ADFS setup. The ADFS configuration wizard was used to specify the Federation Service Name (adfs.task.local) and display name (task.local ADFS). A Group Managed Service Account (GMSA) was created for the ADFS service, and the internal Windows Internal Database (WID) was used. After the setup, all default endpoints (SAML, OAuth, WS-Trust) were verified to be correctly configured and active.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

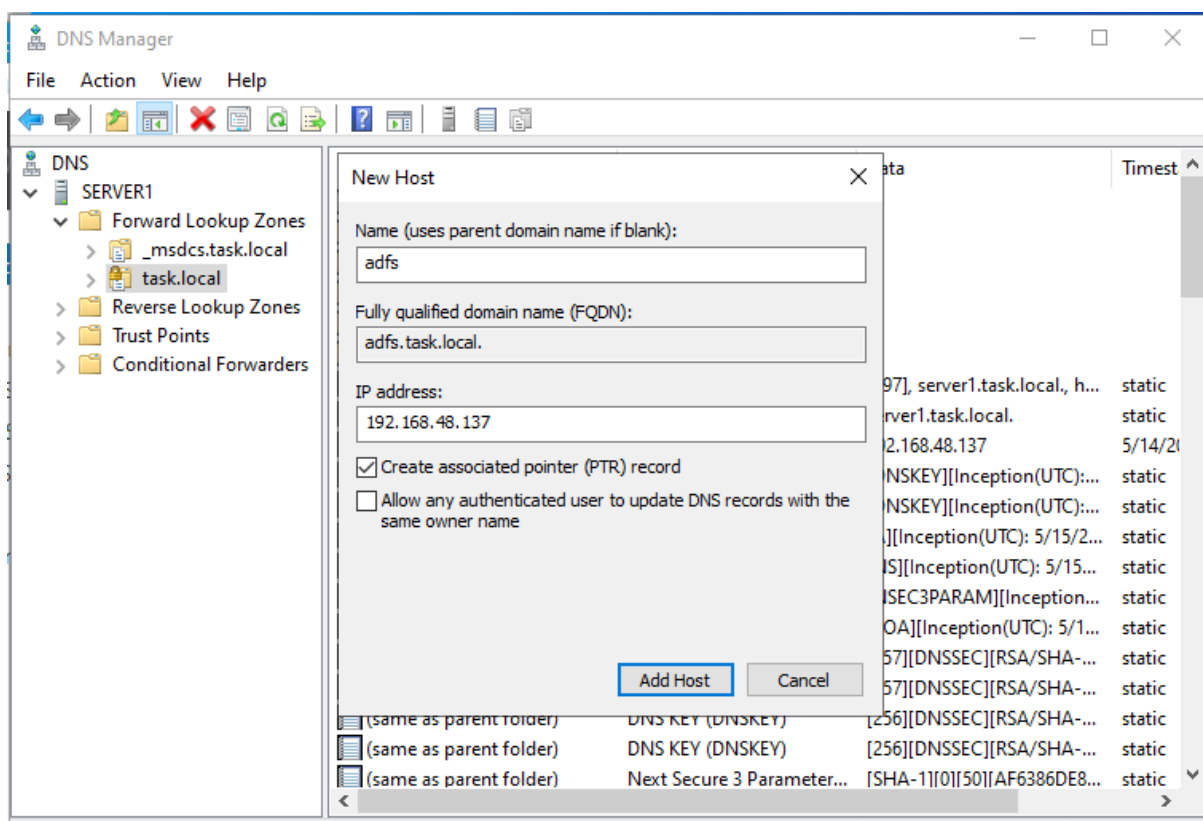
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Install-WindowsFeature ADFS-Federation -IncludeManagementTools

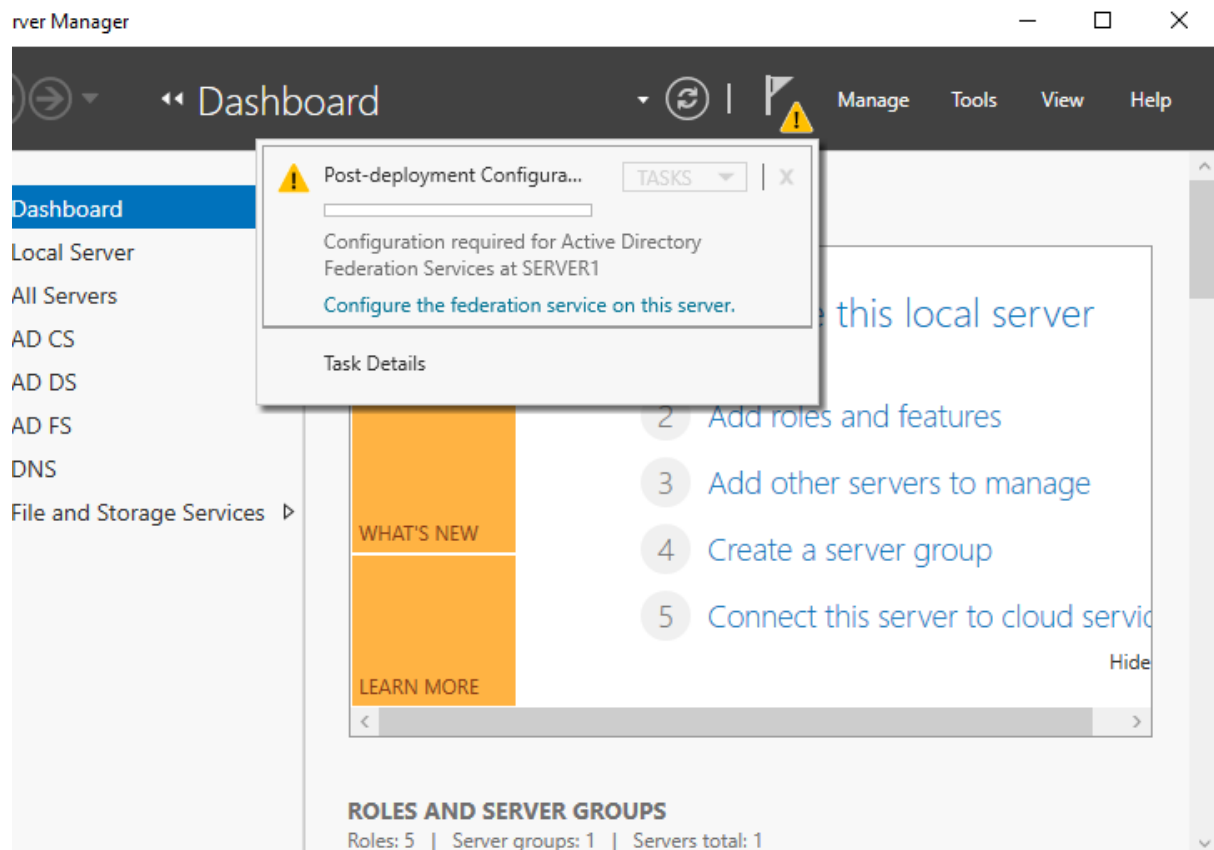
Success Restart Needed Exit Code      Feature Result
-----
True      No          Success      {Active Directory Federation Services}
WARNING: To finish configuring this server for the federation server role using Windows
PowerShell, see http://go.microsoft.com/fwlink/?LinkId=224868.

PS C:\Users\Administrator>
```

(Install ADFS Role)



(DNS resolution for the ADFS service)



(We will now configure Installed ADFS)

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

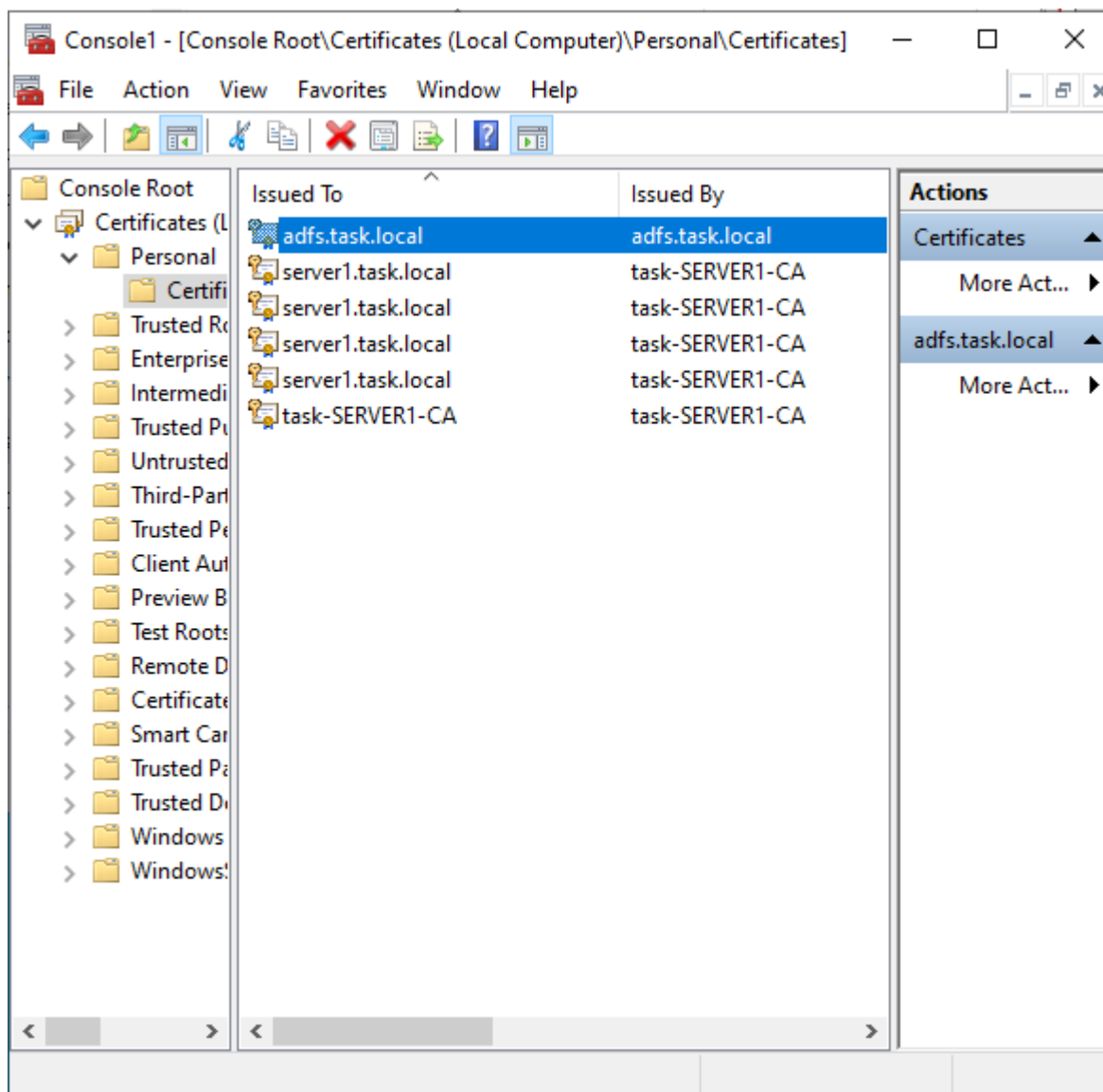
PS C:\Users\Administrator> New-SelfSignedCertificate -DnsName "adfs.task.local" -CertStoreLocation "cert:\LocalMachine\My"

PSParentPath: Microsoft.PowerShell.Security\Certificate::LocalMachine\My

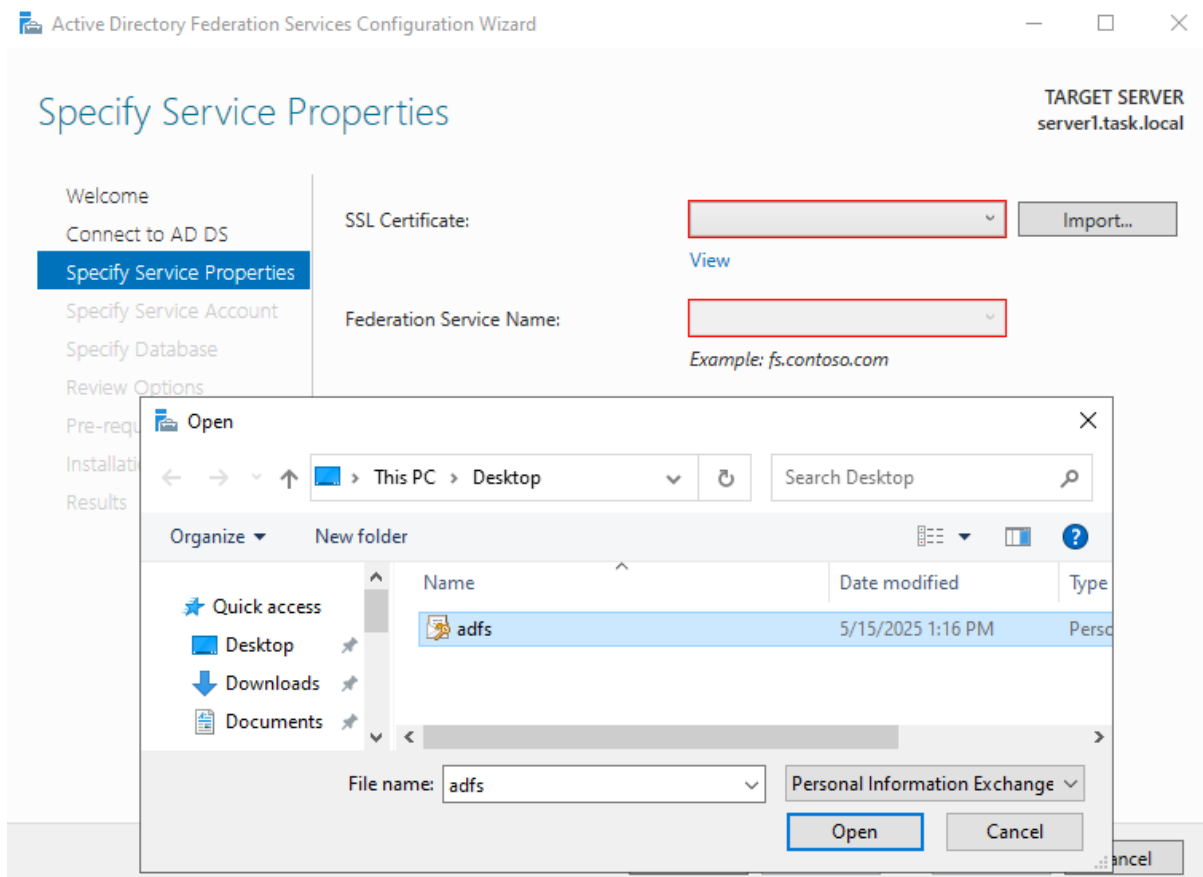
Thumbprint                               Subject
-----
95BB159C0C03560523200CA364342EEC8D246BFA  CN=adfs.task.local

PS C:\Users\Administrator>
```

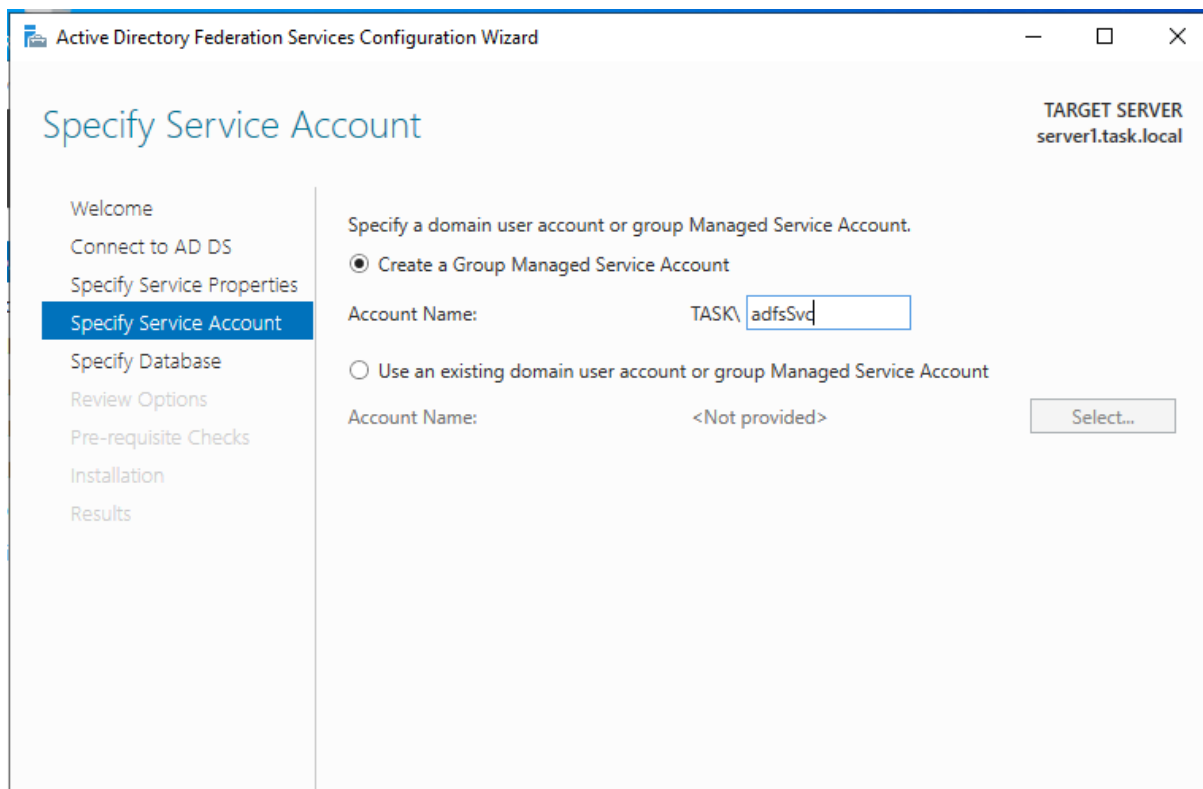
(Generate a ssl certificate)



(Export the certificate)



(Now add our exported certificate)



(Create group managed service account)

Review Options

TARGET SERVER
server1.task.local

Welcome
Connect to AD DS
Specify Service Properties
Specify Service Account
Specify Database
Review Options
Pre-requisite Checks
Installation
Results

Review your selections:

This server will be configured as the primary server in a new AD FS farm 'adfs.task.local'.
AD FS configuration will be stored in Windows Internal Database.
Windows Internal Database feature will be installed on this server if it is not already installed.
A group Managed Service Account TASK\adfsSvc\$ will be created if it does not already exist and this host will be added as a member.
Federation service will be configured to run as TASK\adfsSvc\$.

These settings can be exported to a Windows PowerShell script to automate additional installations

[View script](#)

< Previous

Next >

Configure

Cancel

(review configurations)

Results

TARGET SERVER
server1.task.local

✓ This server was successfully configured

[Show more](#) ×

Welcome
Connect to AD DS
Specify Service Properties
Specify Service Account
Specify Database
Review Options
Pre-requisite Checks
Installation
Results

View detailed operation results

- ⚠ A machine restart is required to complete ADFS service configuration. For more information, see: <https://go.microsoft.com/fwlink/?LinkId=798725>
- ⚠ The SSL certificate subject alternative names do not support host name 'certauth.adfs.task.local'. Configuring certificate authentication binding on port '49443' and hostname 'adfs.task.local'.
- ⚠ The SSL certificate does not contain all UPN suffix values that exist in the enterprise. Users with UPN suffix values not represented in the certificate will not be able to Workplace-Join their devices. For more information, see <http://go.microsoft.com/fwlink/?LinkId=311954>.

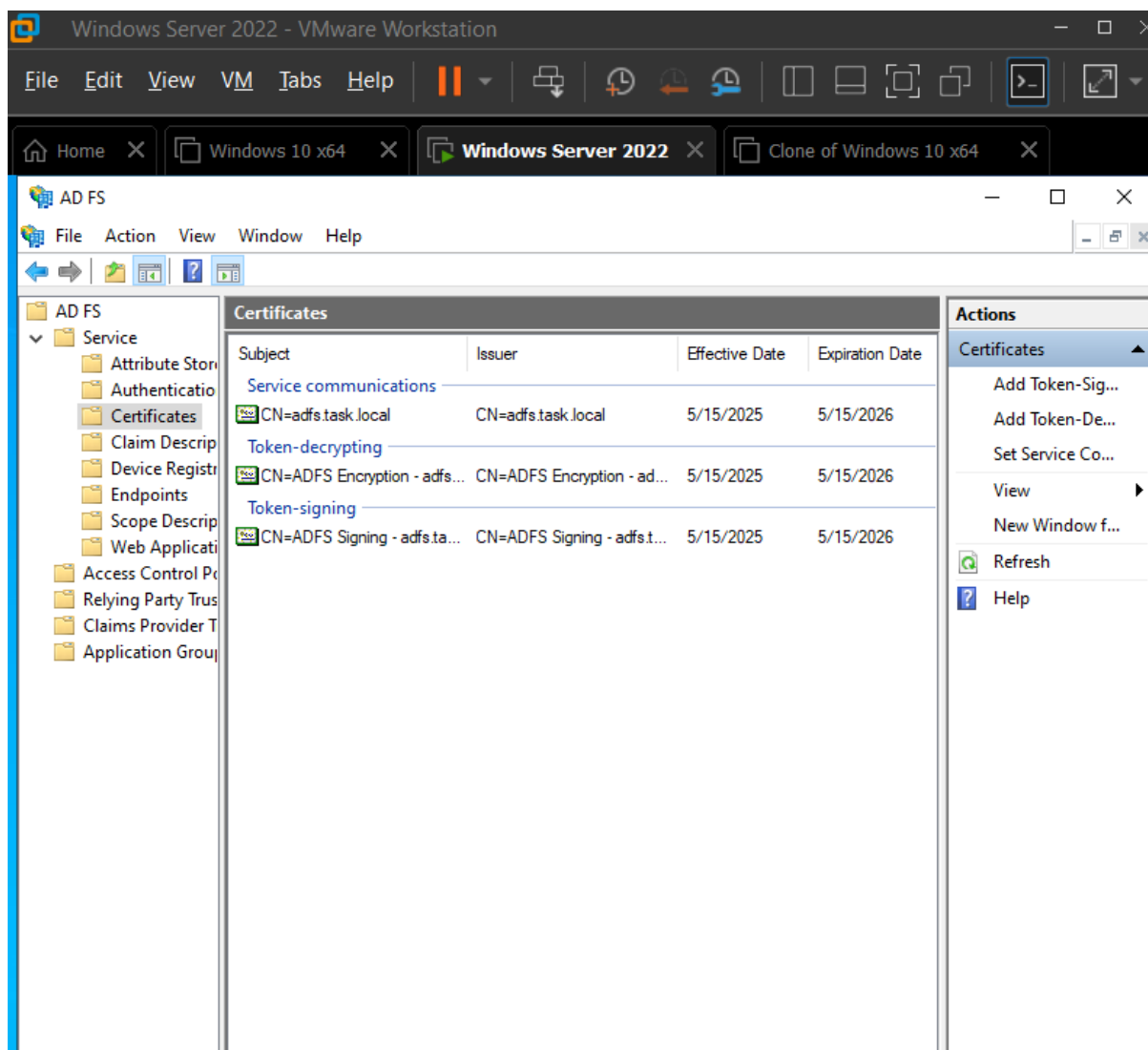
Next steps required for completing your federation service deployment

Need to monitor AD FS service? Use [Azure Active Directory Connect Health](#).

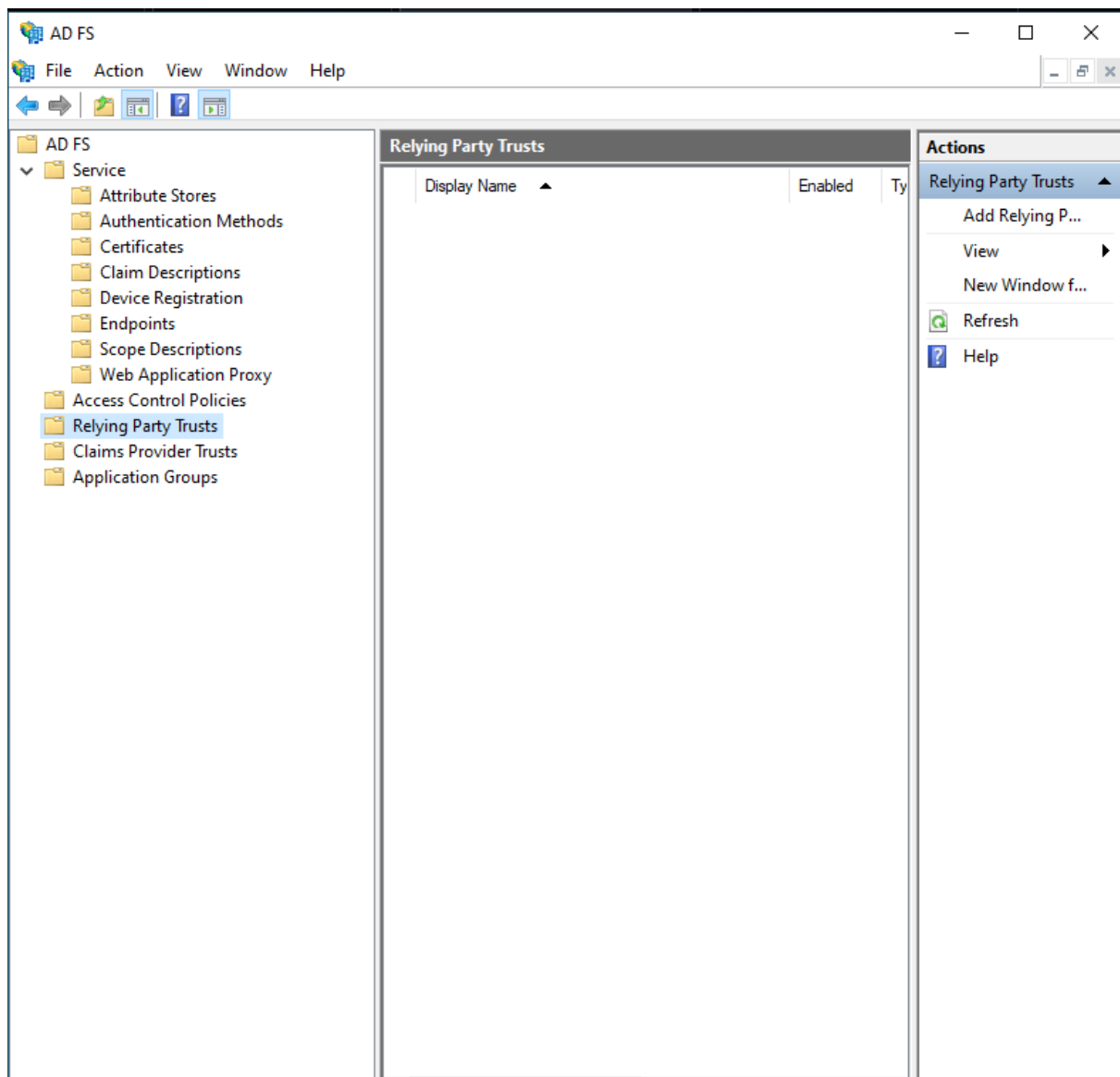
(ADFS configured)

4.2 Monitor Federation Trust Relationships for Anomalies

PowerShell was used to monitor the federation trust configuration. The `Get-AdfsRelyingPartyTrust` cmdlet was executed to list any relying party trusts. As no test applications were integrated, this list remained empty, which was expected. Event Viewer and the `Get-WinEvent` cmdlet were used to review ADFS-related logs. These confirmed that the federation service started without issues and no unexpected behavior was observed. This validated that the federation trust relationship, though minimal in configuration, was functioning correctly.



(Verified service certificates (signing, encryption, service communication))



(Confirmed there are no unknown or misconfigured relying parties in the trust list (it's empty as expected))

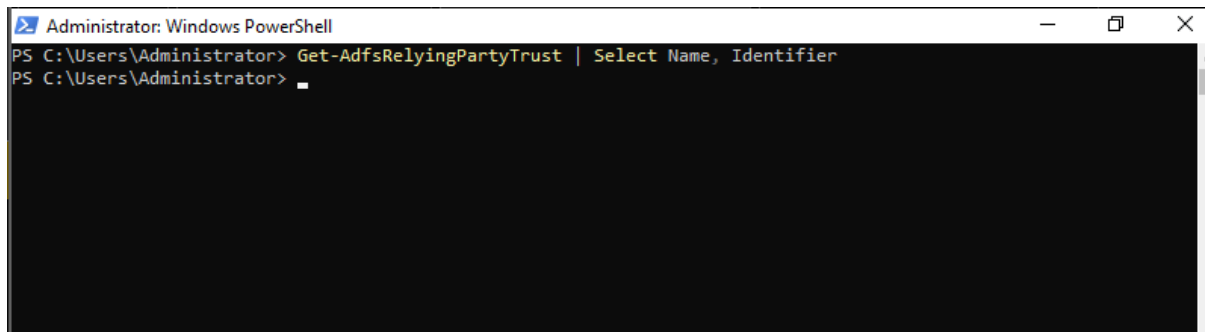
```

PS C:\Users\Administrator> Get-WinEvent -LogName "AD FS/Admin" | Where-Object { $_.Id -in 100, 307, 324, 411 } | Format-List

TimeCreated      : 5/15/2025 1:34:11 PM
ProviderName     : AD FS
Id               : 100
Message          : The Federation Service started successfully. The following service hosts have
                  : been added:
                  : Federation Server Proxy ServiceHost
                  : https://adfs.task.local:443/adfs/services/proxytrustpolicystoretransfer
                  :
                  : MSIS0014: AD FS 1.x Trust Information Service
                  : https://adfs.task.local/adfs/fs/federationserver/service.asmx
                  :
                  : Issuance ServiceHost
                  : http://localhost:80/adfs/services/trust/mexsoap
                  : https://adfs.task.local:443/adfs/services/trust/proxymex/
                  :
                  : Issuance ServiceHost
                  : http://localhost/adfs/services/trust/proxymexsoap
                  : https://adfs.task.local:443/adfs/services/trust/proxymex/
                  :
                  : Issuance ServiceHost
                  : https://adfs.task.local/adfs/services/trust/2005/windowstransport
                  : https://adfs.task.local/adfs/services/trust/2005/certificatemixed
                  : https://adfs.task.local:49443/adfs/services/trust/2005/certificatetransport
                  : https://adfs.task.local/adfs/services/trust/2005/usernamemixed
                  : https://adfs.task.local/adfs/services/trust/2005/kerberosmixed
                  : https://adfs.task.local/adfs/services/trust/2005/issuedtokenmixedasymmetricbasic
                  : 256
                  : https://adfs.task.local/adfs/services/trust/2005/issuedtokenmixedsymmetricbasic2
                  : 56
                  : https://adfs.task.local/adfs/services/trust/13/kerberosmixed
                  : https://adfs.task.local/adfs/services/trust/13/certificatemixed
                  : https://adfs.task.local/adfs/services/trust/13/usernamemixed
                  : https://adfs.task.local/adfs/services/trust/13/issuedtokenmixedasymmetricbasic25
                  : 6
                  : https://adfs.task.local/adfs/services/trust/13/issuedtokenmixedsymmetricbasic256
                  : net.tcp://localhost/adfs/services/trust/tcp/windows
                  :
                  : SAML Metadata
                  : https://adfs.task.local/FederationMetadata/2007-06/
                  :
                  : Other endpoints
                  :
                  : http://+:80/adfs/users/
                  : https://+:443/adfs/oauth2/authorize/
                  : https://+:443/adfs/ls/
                  : https://+:443/adfs/oauth2/logout/
                  : https://+:443/adfs/oauth2/token/
                  : https://+:49443/adfs/oauth2/authorize/
                  : https://+:49443/adfs/ls/
                  : https://+:443/adfs/oauth2/
                  : https://+:443/adfs/oauth2/deviceauth/
                  : http://+:80/adfs/deviceflowresult/
                  : http://+:80/adfs/artifact/

```

(Queried the ADFS Admin event log (Get-WinEvent -LogName "AD FS/Admin") showing Federation Service hosts and endpoints, DNS and service URLs (https://adfs.task.local/...) match configured ADFS endpoint)

A screenshot of a Windows PowerShell window titled "Administrator: Windows PowerShell". The command prompt shows the command `Get-AdfsRelyingPartyTrust | Select Name, Identifier` being executed. The output is not visible in the screenshot, but the command is correctly formatted. The prompt is `PS C:\Users\Administrator>`.

(Checked relying party trusts via `Get-AdfsRelyingPartyTrust` (output confirms no unexpected trusts exist))

5. Recommendations

- Replace the self-signed certificate with one from an internal or trusted external CA.
- Configure relying party trusts with actual applications to simulate SSO behavior.
- Enable regular federation log monitoring to detect anomalies early.
- Set up alerting for failed federation authentication attempts.
- Conduct periodic reviews of ADFS endpoints and trust settings.
- Automate certificate renewal or issue expiration alerts.
- Implement advanced event filtering for Event Viewer to focus on federation-related entries.
- Configure logging retention policies to support long-term audit readiness.
- Restrict access to ADFS Management Console via GPO or local security policies.
- Document the entire federation configuration and trust setup for future audits and reviews.

6. Conclusion

This task established a fully functional Active Directory Federation Services (ADFS) deployment within the task.local domain using Windows Server 2022. The federation service was successfully installed and configured with internal DNS and secured using a self-signed certificate. Federation endpoints were verified for availability, and the service initialization completed without errors. Federation trust relationships were monitored through PowerShell and Event Viewer, confirming that no anomalies were present. Although no Relying Party Trusts were created during this simulation, the environment demonstrated a secure and stable

ADFS deployment along with effective federation health monitoring. This foundation prepares the system for future Single Sign-On (SSO) integrations and enhancements to federation trust configurations